





SQL MURDER MYSTERY

“WHO KILLED THE CEO?”

Major Project for 21 Days SQL Challenge
by Indian Data Club

Investigator: Lead Data Analyst



CASE FILE: THE INCIDENT

BRIEFING

Date: October 15, 2025 | 21:00 Hours

Victim: The CEO of TechNova Inc.

Status: Deceased (Found in Office)

You are the lead data analyst tasked with solving this case. All clues are hidden in the company's databases: Keycard logs, Phone records, and Alibis.

Mission: Find the killer using only SQL queries.

 Detective silhouette analyzing investigation board

🎯 STEP 1: WHERE & WHEN?

We start by identifying the exact location and time of the crime from the evidence logs.

QUERY 01

```
-- Identify where and when the crime happened
SELECT room, found_time
FROM evidence
WHERE room LIKE 'CEO Office'
ORDER BY found_time
LIMIT 1;
```



STEP 2: ACCESS LOGS



Analyzing keycard data to see who entered the critical zone (CEO Office) during the murder window (20:30 - 21:00).

QUERY 02

```
-- Analyze who accessed critical areas at the time
SELECT e.name, e.role, kl.room,
       kl.entry_time, kl.exit_time
FROM keycard_logs kl
INNER JOIN employees e
  ON kl.employee_id = e.employee_id
WHERE kl.room = 'CEO Office'
  AND kl.entry_time BETWEEN
    '2025-10-15 20:30:00'
  AND '2025-10-15 21:00:00'
ORDER BY kl.entry_time;
```


STEP 3: CROSS-CHECKING ALIBIS

We compare claimed locations against actual digital footprints. Any mismatch here indicates a potential suspect.

QUERY 03

```
-- Cross-check alibis with actual logs
SELECT e.name, kl.employee_id,
       kl.room AS actual_location,
       a.claimed_location AS story_location,
       a.claim_time
FROM alibis a
JOIN keycard_logs kl
  ON a.employee_id = kl.employee_id
INNER JOIN employees e
  ON e.employee_id = a.employee_id
WHERE a.claim_time BETWEEN
       kl.entry_time AND kl.exit_time
       AND a.claimed_location != kl.room;
```



STEP 4: SUSPICIOUS ACTIVITY

 Digital network connections data visualization

Tracking communication patterns. Who was making calls during the time of the murder?

QUERY 04

```
-- Investigate suspicious calls
SELECT e.name AS caller,
       e2.name AS receiver,
       c.caller_id, c.receiver_id,
       c.call_time, c.duration_sec
FROM calls c
JOIN employees e
  ON c.caller_id = e.employee_id
JOIN employees e2
  ON c.receiver_id = e2.employee_id
WHERE c.call_time BETWEEN
      '2025-10-15 20:40:00'
      AND '2025-10-15 21:00:00';
```


STEP 5: THE LIE DETECTOR

Using SQL logic to programmatically determine truthfulness.

QUERY 05

```
-- Match evidence with movements and claims
SELECT e.name, e.employee_id,
       a.claimed_location AS what_they_said,
       kl.room AS what_badge_logs_show,
       CASE
         WHEN a.claimed_location = kl.room
         THEN 'Truth'
         ELSE 'LIE DETECTED'
       END AS status
FROM employees e
JOIN alibis a ON e.employee_id = a.employee_id
JOIN keycard_logs kl ON e.employee_id = kl.employee_id
WHERE a.claim_time BETWEEN kl.entry_time
                        AND kl.exit_time
ORDER BY status ASC;
```



STEP 6: CASE CLOSED

 Detective silhouette with final evidence

FINAL VERDICT

By intersecting all data points—location, time, calls, and false alibis—we isolate the perpetrator.

QUERY 06

```
-- Combine all findings to identify the killer
SELECT e.name AS killer
FROM employees e
JOIN keycard_logs kl ON e.employee_id = kl.employee_id
JOIN calls c ON e.employee_id = c.caller_id
JOIN alibis a ON e.employee_id = a.employee_id
WHERE kl.room = 'CEO Office'
      AND c.call_time BETWEEN
          '2025-10-15 20:40:00'
          AND '2025-10-15 21:00:00'
      AND a.claimed_location != kl.room;
```


MISSION ACCOMPLISHED

The killer has been identified.



Data Analyzed



Complex Joins



Justice Served



IMAGE SOURCES



https://images.stockcake.com/public/8/8/6/88678c63-2e5a-4fc0-a046-76948253913a_large/detective-s-investigation-room-stockcake.jpg

Source: stockcake.com



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Source: shop.crimescene.com



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Source: odsc.medium.com



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Source: elements.envato.com



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Source: stockcake.com



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Source: precisionfirm.com

IMAGE SOURCES



<https://i.redd.it/jbq1s3zigdl91.jpg>

Source: www.reddit.com